

EXPERIMENT-14

Implement House Price Prediction using a suitable Machine Learning algorithm.

Code:

```
import pandas as pd
from google.colab import files
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_score
print("=" * 70)
print("          HOUSE PRICE PREDICTION")
print("=" * 70)
uploaded = files.upload()
filename = list(uploaded.keys())[0]
data = pd.read_csv(filename)
print("\nHOUSE PRICE DATASET")
print("-" * 70)
print(data.head(10))
print("\nColumns:")
print(list(data.columns))
# Automatically find price column
price_col = None
for col in data.columns:
    if "price" in col.lower():
        price_col = col
        break
if price_col is None:
    price_col = data.columns[-1]
```

```

print("\nTarget Column :", price_col)

# Use numerical columns
numeric_data = data.select_dtypes(include="number")
X = numeric_data.drop(columns=[price_col])
y = numeric_data[price_col]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)

model = RandomForestRegressor(
    n_estimators=100,
    random_state=42
)

print("\nTraining House Price Prediction Model...")
model.fit(X_train, y_train)
prediction = model.predict(X_test)
print("\nTraining Completed Successfully.")
print("\nActual Prices:")
print(list(y_test[:10]))
print("\nPredicted Prices:")
print([round(x, 2) for x in prediction[:10]])
print("\nR2 Score:")
print(round(r2_score(y_test, prediction), 3))
print("\nMean Squared Error:")
print(round(mean_squared_error(y_test, prediction), 2))
print("\nHouse Price Prediction Completed.")

```

Sample Input:

```

... =====
                        HOUSE PRICE PREDICTION
=====
Choose Files House Pric... Dataset.csv
House Price Prediction Dataset.csv(text/csv) - 91297 bytes, last modified: 8/10/2026 - 100% done
Saving House Price Prediction Dataset.csv to House Price Prediction Dataset.csv

```

Sample Output:

```
... HOUSE PRICE DATASET
...
    Id  Area  Bedrooms  Bathrooms  Floors  YearBuilt  Location  Condition \
0   1  1360         5         4         3     1970  Downtown  Excellent
1   2  4272         5         4         3     1958  Downtown  Excellent
2   3  3592         2         2         3     1938  Downtown    Good
3   4   966         4         2         2     1902  Suburban    Fair
4   5  4926         1         4         2     1975  Downtown    Fair
5   6  3944         1         2         1     1906    Urban    Poor
6   7  3671         1         1         2     1948    Rural    Poor
7   8  3419         2         4         1     1925  Suburban    Good
8   9   630         2         2         1     1932    Rural    Poor
9  10  2185         3         3         1     2000  Downtown    Poor

    Garage  Price
0      No 149919
1      No 424998
2      No 266746
3     Yes 244020
4     Yes 636056
5      No  93262
6     Yes 448722
7     Yes 594893
8     Yes 652878
9      No 340375

Columns:
['Id', 'Area', 'Bedrooms', 'Bathrooms', 'Floors', 'YearBuilt', 'Location', 'Condition', 'Garage', 'Price']

Target Column : Price

Training House Price Prediction Model...

Training Completed Successfully.

Actual Prices:
[514764, 694256, 66375, 650243, 223285, 468127, 513002, 911525, 723265, 339416]

Predicted Prices:
[np.float64(547194.24), np.float64(467334.92), np.float64(470129.22), np.float64(528331.62), np.float64(630166.4), np.float64(693121.92), np.float64(558958.78), np.float64(44

R2 Score:
-0.08

Mean Squared Error:
84010742313.79

House Price Prediction Completed.
```

